

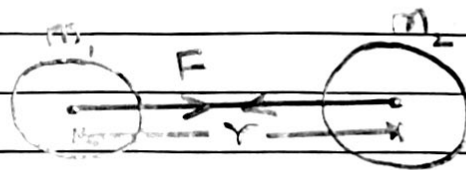
Gravitation

• Newton's law of gravitation

→ It states that, "The gravitational force produced between two bodies is directly proportional to the product of their masses and is inversely proportional to the square of distance between their centres.

i.e. - $F \propto m_1 \cdot m_2$

and $F \propto \frac{1}{r^2}$



Combining these relations we have,

$$F \propto \frac{m_1 \cdot m_2}{r^2}$$

$$\therefore F = \frac{G m_1 \cdot m_2}{r^2} \rightarrow \textcircled{i}$$

G is the universal gravitational constant and its value is $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$

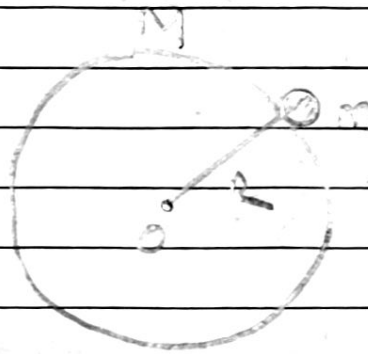
If $m_1 = m_2 = 1 \text{ kg}$ and $r = 1 \text{ m}$ then $F = G$

Hence, Universal gravitational constant is defined as the gravitational force produced between 2 unit masses sep-

erated by unit distance.

• Acceleration due to Gravity (g)

→ The acceleration produced in a freely falling body due to the gravity of earth is called acceleration due to gravity.



Consider a body of mass ' m ' is on the surface of the earth of mass ' M ' and radius ' R ' as shown in figure.

Then, Gravitational Force of the earth on the body is given by.

$$F = \frac{G M m}{R^2} \rightarrow (1)$$

This force gives weight of the body.
 $\therefore F = mg \rightarrow (2)$

From (1) and (2), we have,

$$\frac{GMm}{R^2} = mg$$

$$\Rightarrow \boxed{g = \frac{GM}{R^2}} \longrightarrow (3)$$

Here G and M are constants

$$\boxed{\therefore g \propto \frac{1}{R^2}}$$

This shows that the value of acceleration due to gravity is inversely proportional to the square of radius of the earth.

Since the earth is not perfectly spherical. Its polar radius is less than the equatorial radius. i.e. $R_p < R_e$

So the acceleration due to gravity is more at pole than its equator

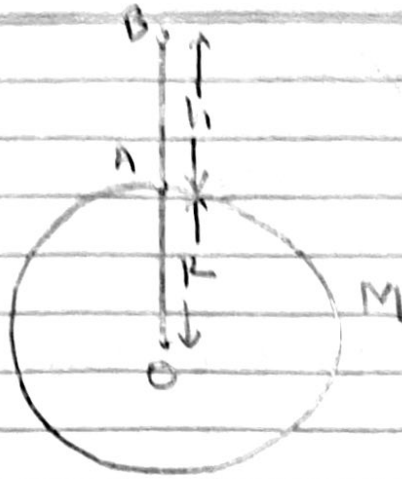
$$\therefore g \text{ at pole} = 9.83 \text{ m/s}^2$$

$$g \text{ at equator} = 9.78 \text{ m/s}^2$$

$$\text{In avg } g = 9.8 \text{ m/s}^2$$

- Variation of acceleration due to gravity with height or Altitude

$\Rightarrow$ Here, The magnitude of acceleration due to gravity on the surface of the earth



th i.e. at A

$$g = \frac{GM}{R^2} \rightarrow (1)$$

Where M is mass and R is radius of earth

Again,

The magnitude of acceleration due to gravity is a height ' h ' from the surface of the earth.

i.e. at B is given by

$$g' = \frac{GM}{(R+h)^2} \rightarrow (2)$$

dividing (2) by (1) we get,

$$\Rightarrow \frac{g'}{g} = \frac{\frac{GM}{(R+h)^2}}{\frac{GM}{R^2}}$$

$$\Rightarrow \frac{g'}{g} = \frac{R^2}{(R+h)^2} = \left(\frac{R}{R+h} \right)^2$$

$$= \left(\frac{R+h}{R+h} \right)^{-2}$$

$$= \left(\frac{R}{R} + \frac{h}{R} \right)^{-2}$$

$$= \left(1 + \frac{h}{R} \right)^{-2}$$

$= 1 - \frac{2h}{R} + \text{terms containing higher power of } \frac{h}{R}$

Here

$\frac{h}{R} < 1$ so ~~neglecting~~ neglecting the

terms containing higher powers of h/R , we get.

$$\boxed{\frac{g'}{g} = 1 - \frac{2h}{R}} \rightarrow \textcircled{3}$$

This shows that the value of acceleration due to gravity decreases with increase in height.

- Variation of acceleration due to gravity with depth.

1



Here, the magnitude of Acceleration due to gravity on the surface of the earth. i.e. At A is given by

$$g = \frac{GM}{R^2} \text{ --- (1) where, } M \text{ is the mass \& } R \text{ is the Radius of Earth.}$$

Again,

The magnitude of acceleration due to gravity at a depth (d) from the surface of the earth is given by.

$$g' = \frac{GM'}{(R-d)^2} \text{ --- (2)}$$

Where M' is the mass of sphere of radius $R-d$.

Dividing (2) by (1), we get

$$\frac{g'}{g} = \frac{\cancel{GM'}}{(R-d)^2} \times \frac{\cancel{GM}}{R^2}$$

$$\Rightarrow \frac{g'}{g} = \frac{M'}{(R-d)^2} \times \frac{R^2}{M} \rightarrow (3)$$

Let ρ be the density of the earth then

$$M = \rho \times V \quad V \rightarrow \text{Volume} \\ = \rho \times \frac{4}{3} \pi R^3$$

And

$$M' = \rho \times \frac{4}{3} \pi V' \\ = \rho \times \frac{4}{3} \pi (R-d)^3$$

Putting these values in eqn (3) we get

$$\frac{g'}{g} = \frac{\cancel{\rho} \times \cancel{\frac{4}{3} \pi} (R-d)^3}{(R-d)^2} \times \frac{R^2}{\cancel{\rho} \times \cancel{\frac{4}{3} \pi} R^3}$$

$$\frac{g'}{g} = \frac{R-d}{R}$$

$$\frac{g'}{g} = \frac{1-d}{R}$$

$$\therefore g' = \left(1 - \frac{d}{R}\right) g \rightarrow (4)$$

This shows that the value of accele-

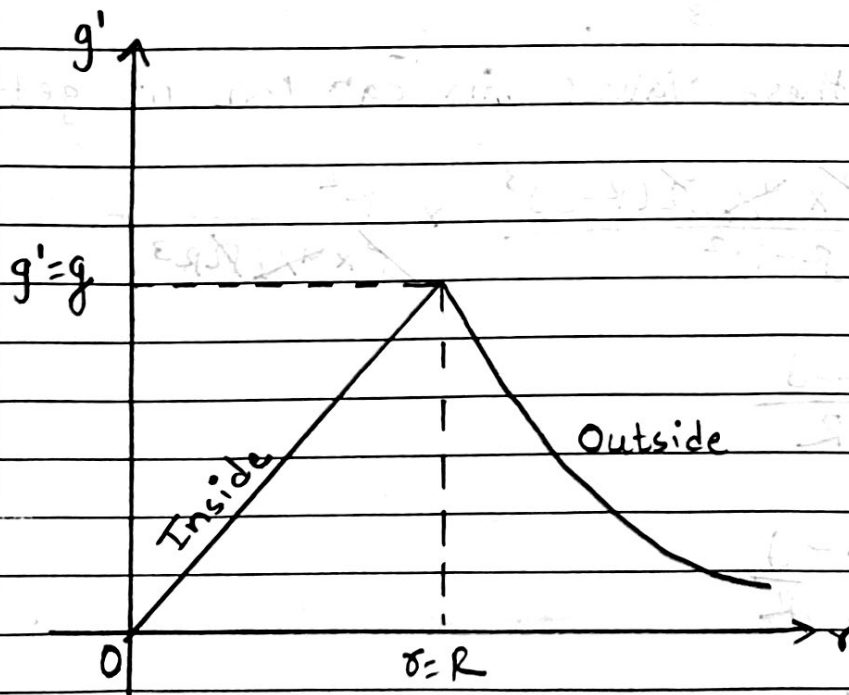
acceleration due to gravity decreases with increase in depth.

At the centre of the earth, $d = R$ then from (i), $g' = \left(1 - \frac{R}{R}\right)g$

$$\therefore g' = 0$$

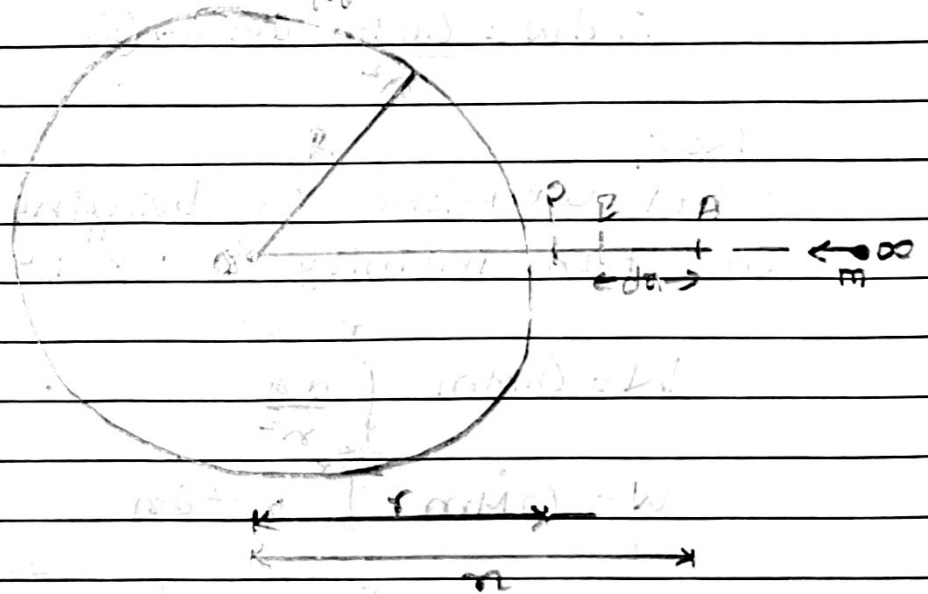
i.e. - The value of acceleration due to gravity at the centre of the earth is 0.

- Graphical representation of Variation of acceleration due to gravity with distance from the centre of the earth.



• Gravitational potential energy

→ The gravitational potential energy at a point in the gravitational field is defined as the amount of work done in bringing a body from infinity to that point.



Consider a point P at a distance 'r' from the centre of the earth of mass 'M' and radius 'R' at which gravitational potential energy is to be determined.

Let A be any position of mass 'm' while bringing it from infinity to P. Such that $OA = R$.

Now,

Gravitational Force of earth on the mass 'm' at A is given by,

$$F = \frac{GMm}{r^2} \quad \text{--- (1)}$$

let,

$AB = dr$ be the small displacement of the mass due to this force. Then small workdone is given by

$$dW = F dr$$

$$\therefore dW = \frac{GMm}{r^2} dr \quad \text{--- (2)}$$

Now,

Total workdone in bringing the mass (m) from infinity to r is given by

$$W = GMm \int_{\infty}^r \frac{dr}{r^2}$$

$$W = GMm \int_{\infty}^r r^{-2} dr$$

$$W = GMm \left[\frac{r^{-2+1}}{-2+1} \right]_{\infty}^r$$

$$W = GMm \left[\frac{r^{-1}}{-1} \right]_{\infty}^r$$

$$W = -GMm [r^{-1} - \infty^{-1}]$$

$$W = -GMm \left[\frac{1}{r} - 0 \right]$$

$$W = -\frac{GMm}{r} \quad \text{--- (3)}$$

This workdone gives gravitational potential energy at a distance ' r ' from the centre

of the earth.

$$\therefore \text{Gravitational potential energy} = -\frac{GMm}{r} \rightarrow (4)$$

• Gravitational Potential

✓ The gravitational potential ϕ at a point in the gravitational field is defined as the amount of work done in bringing a unit mass from infinity to that point.

Note: Derivation is same as above
where $m=1$

$$\text{i.e. } F = -\frac{GM}{r^2} \rightarrow (1)$$

$$dW = F dr$$

$$\therefore dW = \frac{GM}{r^2} dr \rightarrow (2)$$

(fig: same)

$$\therefore W = GM \int_{\infty}^r \frac{dr}{r^2}$$

$$= -\frac{GM}{r}$$

$$\therefore \text{Gravitational potential} = -\frac{GM}{r}$$

On the other hand

Gravitational potential = $\frac{\text{Gr. Potential energy}}{\text{Mass}}$

$$= -\frac{GMm}{r}$$

$$= -\frac{GM}{r}$$

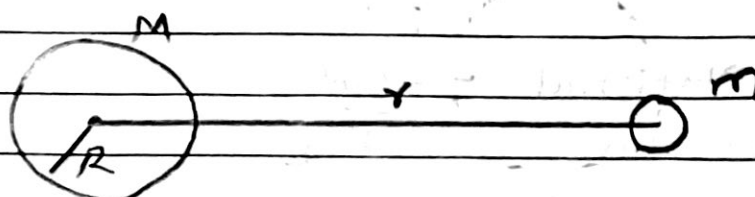
• Gravitational Field

→ The region around the earth upto which gravitational force of the earth can be observed is called Gravitational Field.

• Gravitational Field intensity

→ The Gravitational Field intensity at a point in the gravitational field is defined as the gravitational force experienced by a unit ~~km~~ mass placed at that point.

i.e. $I = \frac{F}{m}$ — (1)



or $I = \frac{GM}{r^2}$

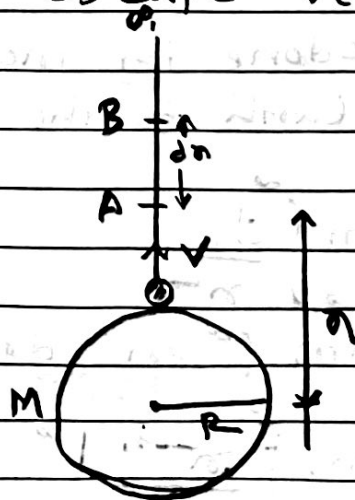
$= \frac{GM}{r^2}$

$\therefore I = \frac{GM}{r^2} = g$

This shows that gravitational Field intensity at a point is numerically equal to the acceleration due to gravity at that point.

• Escape velocity

→ The minimum velocity with which a body is to be projected upward from the surface of earth so that the body just overcomes the gravitational field of the earth and cannot return back to the earth is called escape velocity.



Consider a body of mass 'm' is projected upward with velocity 'v' from the surface of the earth of mass 'M' and radius 'R'.

Then the kinetic energy provided to the body is given by:

$$K.E = \frac{1}{2} mv^2$$

Let A be any position of the body at a distance r from the centre of the earth then gravitational force of earth on the body at the position is given by

$$F = \frac{GMm}{r^2}$$

Let $AB = dr$ be the small displacement of the body then small workdone is given by

$$dW = F dr \\ = \frac{GMm}{r^2} dr$$

Then Total workdone in moving the body from surface of earth to the infinity is given by

$$W = GMm \int_R^{\infty} \frac{dr}{r^2} \\ = GMm \int_R^{\infty} r^{-2} dr \\ = GMm \left[\frac{r^{-2+1}}{-2+1} \right]_R^{\infty}$$

$$\begin{aligned}
 &= GMm \left[\frac{1}{r} - \frac{1}{\infty} \right]_R \\
 &= GMm \left[\frac{1}{R} - \frac{1}{\infty} \right] \\
 &= GMm \left[\frac{1}{R} - 0 \right] \\
 &= \frac{GMm}{R} \rightarrow (2)
 \end{aligned}$$

At escape velocity i.e. $v = v_e$, we have

$$\begin{aligned}
 K.E &= W \\
 \text{or, } \frac{1}{2} m v_e^2 &= \frac{GMm}{R}
 \end{aligned}$$

$$\text{or, } v_e^2 = \frac{2GM}{R}$$

$$\therefore v_e = \sqrt{\frac{2GM}{R}} \rightarrow (3)$$

$$\text{Since, } g = \frac{GM}{R^2}$$

$$\therefore GM = gR^2$$

Then,

$$v_e = \sqrt{\frac{2gR^2}{R}}$$

$$\therefore v_e = \sqrt{2gR} \rightarrow (4)$$

Equation (3) & (4) are the required exp-

Derivation of escape Velocity.

For earth, $g = 9.8 \text{ m/s}^2$ and $R = 6400 \text{ km}$
 $= 6400000 \text{ m}$
 $= 6.4 \times 10^6 \text{ m}$

Then,

$$V_e = \sqrt{2 \times 9.8 \times 6.4 \times 10^6} \text{ m/s}$$

$$= 11200 \text{ m/s}$$

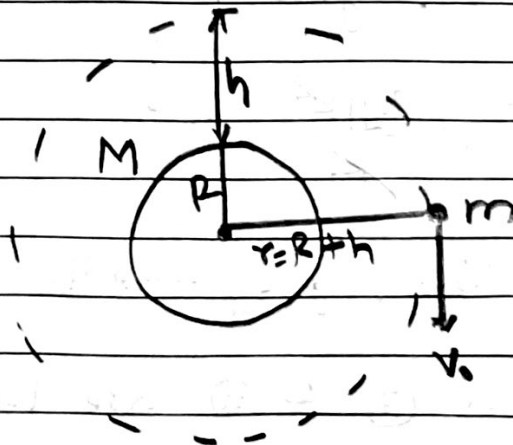
$$= 11.2 \text{ km/s}$$

Q. Why the earth has atmosphere but moon has no atmosphere?

Ans. Velocity of air in earth ~~has~~ is ^{less} ~~more~~ than escape velocity of earth so earth has atmosphere but velocity of air in moon ~~has~~ is more than escape velocity of moon. So moon has no atmosphere.

• Satellite

↳ The heavenly body that revolves around a planet in its own orbit is called satellite.



Consider a satellite of mass ' m ' is revolving around the earth of mass ' M ' and radius ' R ' with orbital velocity ' v_0 ' as shown in figure.

Let ' h ' be the height of satellite then the radius of orbit of satellite is given by

$$r = R + h$$

① orbital velocity

- It is the velocity of satellite in its orbit.

For a satellite, revolving around the earth, the gravitational force between the earth and satellite provides centripetal force to the satellite.

i.e. centripetal force = gravitational force

$$\frac{mv_0^2}{r} = \frac{GMm}{r^2}$$

$$\text{or, } v_0^2 = \frac{GM}{r}$$

$$\therefore v_0 = \sqrt{\frac{GM}{r}} \quad \text{--- ①}$$

Since $g = \frac{GM}{R^2}$

$$\therefore GM = gR^2$$

Then

$$V_0 = \sqrt{\frac{gR^2}{r}} = \sqrt{\frac{gR^2}{R+h}} \rightarrow (2)$$

① Time period

→ It is the time taken by satellite to complete one revolution around the earth.

Since, $V_0 = \omega r$

$$\text{or, } V_0 = \frac{2\pi r}{T}$$

$$\therefore T = \frac{2\pi r}{V_0} \rightarrow (3)$$

Putting the value of V_0 from eqn (2), we get,

$$T = \frac{2\pi r}{\sqrt{\frac{GM}{r}}}$$

$$\text{or, } T = 2\pi r \sqrt{\frac{r}{GM}}$$

$$\therefore T = 2\pi \sqrt{\frac{r^3}{GM}} = 2\pi \sqrt{\frac{(R+h)^3}{GM}} \rightarrow (4)$$

Since $GM = gR^2$

$$\therefore T = 2\pi \sqrt{\frac{(R+h)^3}{gR^2}} \rightarrow (5)$$

(iii) Energy of satellite

→ A satellite revolving around the earth possesses both the kinetic energy and potential energy.

The kinetic energy of satellite is given by,

$$K.E = \frac{1}{2} m v_o^2$$

$$= \frac{1}{2} m \times \frac{GM}{r}$$

$$K.E = \frac{1}{2} \frac{GMm}{r} \rightarrow (6)$$

And, potential energy of satellite is given by,

$$P.E = G \cdot r. \text{ Potential energy}$$

$$= - \frac{GMm}{r}$$

$$\therefore P.E = - \frac{GMm}{r} \rightarrow (7)$$

Then total energy of the satellite is given by

$$\begin{aligned} \text{Total energy} &= K.E + P.E \\ &= \frac{1}{2} \frac{GMm}{r} - \frac{GMm}{r} \end{aligned}$$

$$\therefore \text{Total energy} = - \frac{GMm}{2r}$$

Where negative sign indicates that external energy is needed to remove

the satellite from its orbit. i.e. bounded system.

• Geo-stationary satellite

→ The satellite that appears at rest when observed from the surface of the earth is called geo-stationary satellite.

> The conditions for a satellite to be Geo-stationary are:-

(i) The time period of revolution of satellite should be equal to the time period of rotation of the earth.
i.e. $T = 24$ hours.

(ii) The satellite should revolve in the direction of rotation of the earth.

(iii) The satellite should revolve in the equatorial plane.

• Height of geo-stationary satellite

→ The time period of a satellite is given by $\frac{2\pi}{1} \sqrt{\frac{(R+h)^3}{gR^2}}$

$$\text{or, } T = 2\pi \sqrt{\frac{(R+h)^3}{gR^2}}$$

$$\text{or, } T^2 = \frac{4\pi^2 (R+h)^3}{gR^2}$$

$$c \quad \text{or, } \frac{T^2 g R^2}{4\pi^2} = (R+h)^3$$

$$\text{or, } \left(\frac{T^2 g R^2}{4\pi^2} \right)^{1/3} = R+h$$

$$\therefore h = \left(\frac{T^2 g R^2}{4\pi^2} \right)^{1/3} - R$$

Since, For geo-stationary satellite

$$T = 24 \text{ hrs} = 86400 \text{ s}$$

$$\text{And, } g = 9.8 \text{ m/s}^2$$

$$R = 6400 \text{ km} = 6400 \times 1000 = 6.4 \times 10^6 \text{ m}$$

Then,

$$h = \left[\frac{(86400)^2 \times 9.8 \times (6.4 \times 10^6)^2}{4 \times (3.14)^2} \right]^{1/3} - 6.4 \times 10^6 \text{ m}$$

$$h \approx 36000 \text{ km}$$

• Radius of Parking orbit

$$r = R+h$$

$$= (6400 + 36000) \text{ km}$$

$$= 42400 \text{ km}$$

Homework

- ① Suppose the radius of the earth were to shrink by 2%, its mass remaining the same, would the acceleration due to gravity 'g' increase or decrease and by what percent?

→ Soln,

Let R be the radius of earth, when it is shrunk by 2%. then new radius R' is given as $R' = R - 2\% \text{ of } R$

$$= R - \frac{2R}{100} = R - \frac{R}{50} = \frac{49}{50} R$$

The acceleration due to gravity of earth is, $g = \frac{GM}{R^2}$, where M = mass of earth and G

G = Universal gravitational constant.

When R is shrink by 2%, then new g' is

$$g' = \frac{GM}{R'^2} = \frac{GM}{\left(\frac{49}{50} R\right)^2} = \frac{GM}{R^2} \times \left(\frac{50}{49}\right)^2 = g \times 1.041$$

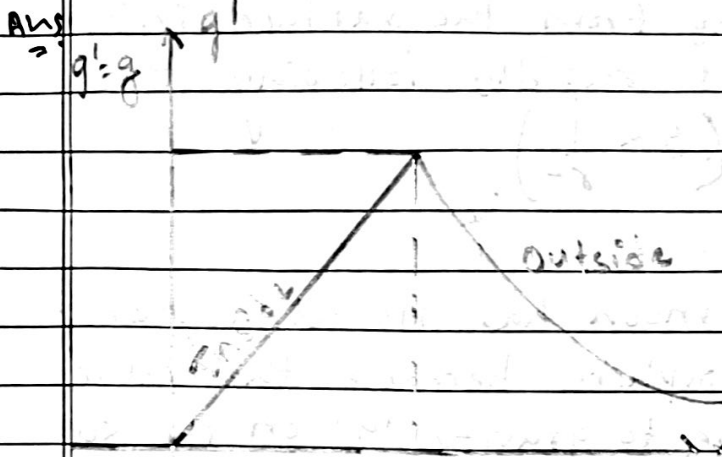
$$\therefore \text{Change} = \frac{g' - g}{g} \times 100\%$$

$$= \frac{1.041g - g}{g} \times 100\%$$

$$= 4.1\%$$

$\therefore$ The acceleration due to gravity is changed by 4.1%.

- ② How does ' g ' at a point vary with the distance from the centre of the earth? Where is the highest value of g ? Explain with graph.



The acceleration due to gravity (g) is not constant and varies depending on the distance (r) from the centre of the earth (R).

- At centre ($r=0$): The value of ' g ' is zero because the mass of earth surrounds the point equally in all directions, canceling out the gravitational pull.
- Inside the earth ($r < R$)
 - As we move from the centre toward the surface, ' g ' increases linearly with distance.
- At surface ($r=R$)
 - The value of ' g ' reaches its maximum point

because it is the closest possible distance to the entire mass of the earth.

- Outside the Earth ($r > R$)

- As we move away from the surface into space, 'g' decreases rapidly, following the inverse square law ($g \propto \frac{1}{r^2}$).

③ Explain why the moon has no atmosphere?

Ans Moon has no atmosphere because the value of acceleration due to gravity 'g' on the surface of moon is small. Therefore the value of escape velocity on the surface of moon is small (about 2.5 km/s). The value of r.m.s velocity of the molecule of different gases is much above the value of escape velocity on moon. That is why all the molecule of gases have escaped and there is no atmosphere on moon.

④ A person sitting in an artificial satellite of the earth feels weightlessness, but a person standing on the moon has weight though the moon is a satellite of the earth. Why?

Ans An astronaut inside a small space ship orbiting around the earth does not experience any gravity. But an astronaut on the moon which is also orbiting around the earth does experience gravity because force

of gravity is given as,

$$F = \frac{GM}{r^2}$$

The mass of spaceship is very small compared to the moon. Due to this reason, the force of gravity on spaceship is negligible compared to that of moon.

- ⑤ An astronaut releases a spoon out of a satellite in the space. Will the spoon fall on the earth?

Ans. No, the spoon will not fall on the earth when released from a satellite in the space. The gravitational force provided by earth is spent to produce centripetal force and no net force acting on it. So, it moves around the orbit of satellite with uniform speed.

Numerical and Figure based Questions

- 1) Mass of earth is nearly 81 times heavier than the moon. Its diameter is about 4 times larger than that of the moon. Estimate the value of acceleration of gravity on the surface of the moon.

→ Soln,

$$M_e = 81 M_m \text{ (let)}$$

$$M_m = \frac{1}{81} M_e$$

$$R_e = 4 R_m \text{ [Since diameter is 4 times larger]}$$

$$\Rightarrow R_m = \frac{R_e}{4}$$

We know that,

$$g = \frac{GM_e}{R_e^2} = 9.8 \rightarrow (i)$$

Now,

For moon,

$$g' = \frac{GM_m}{R_m^2}$$

$$= \frac{G \times \frac{1}{81} M_e}{\left(\frac{R_e}{4}\right)^2}$$

$$= \frac{G \times 16 M_e}{81 R_e^2}$$

$$= \frac{16}{81} \times \frac{GM_e}{R_e^2}$$

$$= \frac{16}{81} \times 9.8$$

$$= 1.94 \text{ m/s}^2$$

- 2) At what altitude, above the earth's surface, would the acceleration due to gravity be 5 m/s^2 . [Earth's radius = 6400 km , acceleration due to gravity on the surface of the earth = 10 m/s^2].

→ Soln,

$$g = 10 \text{ m/s}^2$$

$$R = 6400 \text{ km} = 6.4 \times 10^6 \text{ m}$$

$$g' = 5 \text{ m/s}^2$$

$$h = ?$$

We know that,

$$g = \frac{GM}{R^2}$$

$$10 = \frac{GM}{(6.4 \times 10^6)^2}$$

$$10(6.4 \times 10^6)^2 = GM \rightarrow (i)$$

$$g' = \frac{GM}{(R+h)^2}$$

$$5 = \frac{GM}{(6.4 \times 10^6 + h)^2} \rightarrow (ii)$$

dividing (i) by (ii) we get,

$$\frac{10(6.4 \times 10^6)^2}{5} = \frac{GM}{(6.4 \times 10^6 + h)^2}$$

$$\Rightarrow 8.192 \times 10^{13} = (6.4 \times 10^6 + h)^2$$

$$\Rightarrow (8.192 \times 10^{13})^{1/2} = 6.4 \times 10^6 + h$$

$$\Rightarrow 9050966.799 = 6.4 \times 10^6 + h$$

$$\Rightarrow h = 9050966.799 - 6.4 \times 10^6$$

$$h = 2650966.799 \text{ m}$$

$$\therefore h = 2650 \text{ km}$$

3) Calculate the points along a line joining the centers of earth and moon where there is no gravitational force.

→ Soln,

$$M_e = 6 \times 10^{24} \text{ kg}$$

$$M_m = 7.4 \times 10^{22} \text{ kg}$$

$$d = 3.8 \times 10^8 \text{ m}$$

We know that,

At neutral point, the gravitational field strength (g) from both bodies is equal.

$$\frac{GM_e}{r^2} = \frac{GM_m}{(d-r)^2}$$

$$\text{or, } \frac{6 \times 10^{24}}{r^2} = \frac{7.4 \times 10^{22} \text{ kg}}{(3.8 \times 10^8 - r)^2}$$

$$\text{or, } 81.08 = \left(\frac{r}{3.8 \times 10^8 - r} \right)^2$$

$$\text{or, } (81.08)^{1/2} = \frac{r}{3.8 \times 10^8 - r}$$

$$\text{or, } 9.004 = \frac{r}{3.8 \times 10^8 - r}$$

$$\text{or, } 3421711284 - 9.004r = r$$

$$\text{or, } 10.0045r = 3421711284$$

$$\text{or, } r = \frac{3421711284}{10.0045}$$

$$\text{or, } r = 3.42 \times 10^8$$

∴ At $3.42 \times 10^8 \text{ m}$ from the centre of earth there is no gravitational force,

4) A - remote sensing satellite of the earth revolves in a circular orbit at a height of 250 km above the earth's surface. What is the orbital speed and period of revolution of the satellite?

→ Given,

$$R = 6400 \text{ km} = 6.4 \times 10^6 \text{ m}$$

$$h = 250 \text{ km} = 250,000 \text{ m}$$

$$V_0 = \sqrt{\frac{gR^2}{R+h}}$$

$$= \sqrt{\frac{(9.8) (6.4 \times 10^6)^2}{6.4 \times 10^6 + 2.5 \times 10^5}}$$

$$= \sqrt{\frac{4.01408 \times 10^{14}}{6650000}}$$

$$= \sqrt{60362.09}$$

$$\therefore V_0 = 7769.3 \text{ m/s}$$

$$T = \frac{2\pi R}{V_0}$$

$$= \frac{2 \times 3.14 \times 6650000}{7769.3}$$

$$= 5377.98$$

$$\therefore T = 5377.98 \text{ sec}$$

5) An artificial satellite revolves around the earth in 2.5 hours in a circular orbit. Find the height of the satellite above the earth assuming earth as a sphere of radius 6370 km.

→ Soln,

$$T = 2.5 \text{ hrs} = 9000 \text{ sec}$$

$$R = 6370 \text{ km} = 6.37 \times 10^6 \text{ m}$$

$$T = 2\pi \sqrt{\frac{(R+h)^3}{gR^2}}$$

$$\text{or, } 9000 = 2\pi \sqrt{\frac{(6.37 \times 10^6 + h)^3}{9.8 \times (6.37 \times 10^6)^2}}$$

$$\text{or, } (1432.39)^2 = \frac{(6.37 \times 10^6 + h)^3}{3.97 \times 10^{14}}$$

$$\text{or, } 8.14 \times 10^{20} = (6.37 \times 10^6 + h)^3$$

$$\text{or, } 9339105.102 = 6.37 \times 10^6 + h$$

$$\therefore h = 2969105.102 \text{ m}$$

6) Obtain the value of g on earth's surface from the motion of moon assuming that its period of rotation round the earth is 27 days ~~at~~ 8 hours and the radius of its orbit is 60 ~~at~~ times the radius of the earth.

→ Given,

$$T = 27 \text{ days } 8 \text{ hours}$$

$$= (27 \times 86400 + 8 \times 3600) \text{ Sec}$$

$$= 2361600 \text{ sec}$$

$$r = 60.1 R$$

Now,

$$T = 2\pi \sqrt{\frac{r}{GM}}$$

$$\text{or, } 2361600 = 2\pi (60.1 R) \sqrt{\frac{60.1 R}{gR^2}}$$

$$\text{or, } 6253.91 = R \sqrt{\frac{60.1}{gR}}$$

$$\text{or } 39111457.43 = R^2 \left(\frac{60.1}{9R} \right)$$

$$\text{or } \frac{R}{8} = 650773.0021$$

$$\text{or } g = \frac{6.4 \times 10^6}{650773.0021}$$

$$\therefore g = 9.83 \text{ m/s}^2$$

7) A 200 kg satellite is lifted to an orbit of $2.2 \times 10^4 \text{ km}$ radius. If the radius and mass of the earth are $6.37 \times 10^6 \text{ m}$, and $5.98 \times 10^{24} \text{ kg}$ respectively, how much additional potential energy is required to lift the satellite?

→ Given,

$$m = 200 \text{ kg}$$

$$R = 6.37 \times 10^6 \text{ m}$$

$$r = 2.2 \times 10^4 \times 1000 \text{ m} = 2.2 \times 10^7 \text{ m}$$

$$M = 5.98 \times 10^{24} \text{ kg}$$

Initial potential energy of satellite is,

$$PE_i = -\frac{GMm}{R}$$

$$= -\frac{6.67 \times 10^{-11} \times 5.98 \times 10^{24} \times 200}{6.37 \times 10^6}$$

$$= -1.25 \times 10^{10}$$

Final potential energy of satellite is

$$PE_f = \frac{-GMm}{r}$$

$$= \frac{-6.67 \times 10^{-11} \times 5.98 \times 10^{24} \times 200}{2.2 \times 10^7}$$

$$= -3.62 \times 10^9$$

$$\text{Additional PE} = PE_f - P.E_i$$

$$= -3.62 \times 10^9 + 1.25 \times 10^{10}$$

$$\therefore \text{Additional PE} = 8.87 \times 10^9 \text{ Joules}$$

⑧ Communications satellites are usually placed in a geo-synchronous orbit. Given that the mass of the earth is $6 \times 10^{24} \text{ kg}$ and its mean radius is $6.4 \times 10^6 \text{ m}$.

i) Show that the radius of a geo-synchronous orbit must be $4.28 \times 10^7 \text{ m}$.

ii) Calculate the increase in potential energy of a satellite of mass 750 kg when it is raised from the earth's surface into a geo-synchronous orbit.

→ Given,

$$M_e = 6.0 \times 10^{24} \text{ kg}$$

$$R = 6.4 \times 10^6 \text{ m}$$

① To show $r = 4.28 \times 10^7 \text{ m}$

$$T = 2\pi \sqrt{\frac{r^3}{GM}}$$

$$\text{or, } T = 2 \times (3.14) \sqrt{\frac{r^3}{6.67 \times 10^{-11} \times 6.04 \times 10^{24}}}$$

$$\text{or, } 24 \times 3600 = 6.28 \times \sqrt{\frac{r^3}{40.28 \times 10^{13}}}$$

$$\text{or, } 7464960000 = 39.43 \times \frac{r^3}{40.28 \times 10^{13}}$$

$$\text{or, } r^3 = 7.62 \times 10^{22}$$

$$\text{or, } r = 4.28 \times 10^7$$

$$\therefore r \approx 4.28 \times 10^7$$

Proved

ii)

→ Increase in potential energy

$$\Delta U = U_f - U_i$$

$$= \left(-\frac{GMm}{r} \right) - \left(-\frac{GMm}{R} \right)$$

$$\Delta U = GMm \left(\frac{1}{R} - \frac{1}{r} \right)$$

$$= (6.67 \times 10^{-11}) \times (6.0 \times 10^{24}) \times 750 \left(\frac{1}{6.4 \times 10^6} - \frac{1}{4.28 \times 10^7} \right)$$

$$\approx 3.0 \times 10^{14} (1.56 \times 10^{-7} - 0.23 \times 10^{-7})$$

$$\therefore \Delta U \approx 3.99 \times 10^{10} \text{ Joules}$$